SRM IST DELHI NCR CAMPUS, MODINAGAR

DEPARTMENT OF COMPUTER SCIENECE AND ENGINERING

DATA SCIENCE (21CSS303T)

3ND YEAR,6TH SEMESTER

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ASSIGNMENT-4

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SUBMITTED TO:

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Q1> Explain the concept of data wrangling and its importance.

Ans It is a process of organizing, cleaning and transforming raw data into a structured format so it can be used for analysis. In real world scenarios, raw data collected from different sources is often incomplete, inconsistance or unorganized.

Importance of Data Wrangling:

1. Improve the quality and reliability of data
2. Helps in removing errors, inconsistant and duplicates
3. Makes data structure and ready for analysis
4. Saves time during analysis and modeling

In data science a large portion of time is spent on data wrangling high-quality data leads to better insights and predictions.

Q2> Explain string manipulation in python with suitable code ~~exp~~ examples.

Ans String manipulation refers to the process of modifying, analyzing or processing text data using various string function available in python. Since many datasets contain text information such as names, addresses or messages, strings manpulation is an important part of data processing.

Python provides many building methods for working with strings such as changing case, searching, replacing, slicing and conatenation.

<code>

```
text = "Hello World"
~~plo~~ print(text.lower())
print(text.upper())
print(text.replace("World","python"))
```

...(text [0 5])

```
name = "Lakshmi"
greeting = "Hellow" + name
print (greeting)
```

String manipulation helps in cleaning and processing textural data before analysis.

Q3> Explain the general techniques and programming practices used for efficient handling of large datasets in data science.

Ans Handling large dataset efficiently is an important challenge in data science because large amount of data require more memory, processing power and time. Therefore, several techniques and practices are used to process large datasets effectively.

Common techniques:

1. Data Sampling : Analyze a small representative portion of data.
2. Chunk Processing : Read data in small parts instead of ~~location~~ loading the entire dataset.
3. Using Efficient Libraries : Tools like NumPy and Pandas for faster processing.
4. Parallel Processing: Use multiple CPU cores to process data faster.
5. Data Compression : Reduce storage and memory usage.

These techniques help in improving performance, reducing processing time, and managing large scale data effectively.

Q4> Explain different data transformation techniques used in preprocessing

Ans Data transformation is a preprocessing step where raw data is converted into a suitable format for analysis or machine learning algorithm. This step ensure that the data is consistent, standardized and meaningful.

Common data transformation techniques include :

1. Normalization : Scaling values b/w 0 & 1 .
2. Standardization : Adjusting data to have mean 0 and standard deviation 1 .
3. Encoding : Converting categorical data into numeric form .
4. Aggregation : Combining multiple values into summery statistics .
5. Smoothing : Removing noise from data .